Earnings Growth, Job Flows and Churn

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March, 2020

Abstract

How much do workers making job-to-job transitions benefit from moving away from a shrinking and towards a growing firm? We show that earnings growth in the transition increases with net employment growth at the destination firm and, to a lesser extent, decreases if the origin firm is shrinking. So, we sum the effect of leaving a shrinking and entering a growing firm and remove the excess turnover-related hires because gross hiring has a much smaller association with earnings growth than net employment growth. We find that job-to-job transitions with the cross-firm job flow have 23% more earnings growth than average.

*University of Queensland; U.S. Census Bureau, CES; Stony Brook University. Thanks to our discussant Isabel Cairo and to Mark Bils, Jason Faberman, Henry Hyatt, Leo Kaas, Narayana Kocherlakota, Guido Menzio, Ryan Michaels, Aysegul Sahin, Steven Stern and Ronald Wolthoff. This research uses data from the Census Bureau's Longitudinal Employer Household Dynamics Program, which was partially supported by the following National Science Foundation Grants SES-9978093, SES-0339191 and ITR-0427889; National Institute on Aging Grant AG018854; and grants from the Alfred P. Sloan Foundation. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 1819. All results have been reviewed to ensure that no confidential information is disclosed. All errors are our own.

Workers' earnings changes are concentrated around periods in which they change jobs (see e.g. Topel and Ward (1992); Moscarini and Postel-Vinay (2017); Hahn et al. (2017)). From a different perspective, workers job changes are part of the flow of jobs from firm to firm, reallocating productive factors and, presumably increasing productivity and creating a surplus (e.g. Hsieh and Klenow (2009); Hagedorn et al. (2017)). This paper asks how the two relate, specifically, how much does a worker gain by being part of a job flow. Implicit in this question is a hypothesis that some worker-level transitions are different than others, as some reallocate labor from shrinking firms to growing firms and others are part of the excess churn. Then, is the promise of earnings growth what induces a worker to be part of the job flow, or from the other side, is some of the dividend from job reallocation shared with the worker?

In this paper, we show that labor-force dynamics at both destination and origin firms influence the earnings outcomes of workers. This is to say that, even beyond the firm's fixed characteristics such as age, size, or whether they are high or low average wage firms, the labor-force dynamics at these firms affects workers' earnings growth. These employment dynamics include net employment growth as well as gross flows, i.e. churn or excess hiring. Quantitatively, these labor-force dynamics are important to earnings growth in a job transition: in a regression, the elasticity of earnings growth to the destination firms' employment growth is about half the size of the elasticity with respect to the firms' average wage level. Differences in the growth rates at the destination and origin firm imply a 90-10 differential in earnings growth of \$1,624, which is about 20% of the 90-10 differential implied by the difference in average wages at origin and destination. In other words, workers' earnings increase when they move to a faster-growing firm, and this is quantitatively important even after controlling for other firm characteristics.

Not every worker hired at a growing firm, however, directly accounts for its net growth because these firms are simultaneously separating and replacing workers and, similarly a separation from a shrinking firm may be replaced. In fact, we document that firms with faster net growth have even higher separation rates and thus larger gross flows. Faster shrinking firms also have higher hiring rates and higher gross flows than a more slowly shrinking firm. These patterns are important to our central question because there is an empirical distinction between net employment growth, to which earnings growth has a large elasticity, and gross flows, to which earnings are less sensitive.

To answer how a worker's earnings growth is affected by being part of the cross-firm job flow, a job reallocation, is composed of three factors: the positive effect from being hired at a growing firm, the negative effect from leaving a shrinking firm and both are tempered by the less-than-one probability this worker's transition contributed to the net employment change at the origin or destination firm.

Conditional on a bevy of other characteristics, the earnings-growth effect of being part of a job reallocation we find to be 1.7 percentage points. Given that the average earnings gain in our sample is about 7.3%, the premium associated with being part of the job flow is 23% of the expected gains from a job-to-job transition. Because the probability that a job transition is part of a job flow is about 21%, transitions with the job flow account for about 6% of overall earnings growth in job-to-job transitions.

This project links the hiring and separation rates of firms to workers' job-to-job transitions. To do so, we use the Longitudinal Employer-Household Dynamics (LEHD) dataset, administrative data that links nearly every worker to their firm and measures earnings at a quarterly frequency. From this dataset, we pull nearly 2.7 million job-to-job transitions from 1998-2013 and drawn from 17 states. Because this is administrative data and relatively high frequency, this dataset lets us identify job-to-job transitions and gives us a clear measure of workers' earnings growth from the surrounding periods. This relatively long panel also allows us to see multiple job transitions, meaning we can use individual-level fixed effects in our estimates to control for worker composition reflecting, e.g. sorting. More fundamentally, our question inherently requires matched employee-employer data because worker-side survey data does not observe changes at the firm-level.

These estimates of the job-flow earnings premium suggest that wages are an important signal for reallocating workers towards growing firms, similar to Carrillo-Tudela et al. (2020). These relatively high wages may reflect the recruiting process of firms that wish to grow—higher wages increase the yield on a vacancy—or they may result from rent sharing because growing firms pass along some of the productivity shock that caused them to grow. Our results suggest an important discipline on these two motives: firm growth, not hiring itself, has a stronger effect on earnings growth for workers. For one studying firm dynamics, these results inform the costs of reallocating inputs—growing firms pay extra to increase their stock of labor. To understand workers' earnings dynamics, these results highlight one of the reasons that job transitions may have very disparate outcomes: firm dynamics influence the potential gains.

We are not the first to have looked at earnings growth across job-to-job transitions. Topel and Ward (1992) is an important early example emphasizing the outsized role of job transitions in overall earnings growth. Hahn et al. (2017) further decomposes changes in aggregate earnings due to entrants, exits, transitions, and stayers. Relating job transitions to the characteristics of firms, Haltiwanger et al. (2017) highlight the reallocative role of job-to-job transitions from less productive to more productive firms and Bachmann et al. (2017) and Elsby et al. (2017) are important examples considering the joint-dynamics of gross and net flows of a firm. While the previous three papers focus solely on flows, our paper analyzes the effects of these dynamic characteristics on earnings growth. The role of firm characteristics such as age, size, and average wages on earnings has focused on the static characteristics of the firm and on workers' wage levels. Our paper extends these findings to show that these same static and dynamic firm characteristics affect the growth of workers' earnings.

These facts about earnings growth in job transitions and our focus on employer dynamics serve as evidence to the role of the firm in a frictional labor market and should help to inform a growing literature of labor market search models with large firms. Belzil (2000) establishes a relationship between faster-growing firms and wages above the workers' fixed effect. Davis et al. (2013) find that vacancy-fill rates are related to firm-level characteristics such as employment growth, turnover, and size. Similarly, Kettemann et al. (2018) show that job-filling rates are higher at high-paying firms and that these rates are increasing in firms growth rates even after controlling for the firm's static components of wages. Our results suggest that employer growth and gross worker flows are not just important for job-filling rates, but also for the earnings growth of workers. Earnings growth, instead of levels, is an important distinction between Kettemann et al. (2018) and ourselves: by focusing on the growth rate, we difference out unobservable individual-level characteristics which may not be randomly allocated across destination firms.

A number of structural models of firm recruiting are consistent with our empirical findings. Our finding that growing firms pay a wage premium suggest that in addition to the increased recruitment intensity in Gavazza et al. (2018) or Bilal et al. (2019), firms use higher wages to recruit workers. One example of such a framework that is consistent with our earnings result is convex recruitment costs and directed search found in Kaas and Kircher (2015).

The rest of the draft is organized as follows. In Section 1 we describe the LEHD and the variables we build in it. In Section 2 we present the unconditional relationships between earnings growth, job flows and churn. In Section 3 we describe our findings from an earnings change regression. Then, in Section 4, we construct the probability that a worker flow is part of a job flow, and quantify the associated earnings growth premium. Finally, we conclude in Section 5.

1 The Data

Our data comes from the LEHD, a matched employer-employee panel put together from administrative data collected by states to provide unemployment insurance files. Because states begin participating at different times, we trade off geographic breadth and time-frame. We opt for a 17 state sample, which includes Texas and California and runs from 1998-2014. The observations are quarterly and we observe only earnings, rather than the hourly wage.

1.1 Sample Selection

To construct our sample, we randomly draw 5% of individuals in the labor force in our 17 states.[1] Employment in these states represents 42% of national employment. Because the dataset is created from unemployment accounts, we do not include anyone not covered by unemployment insurance, which includes all federal employees. Also because of this provenance, the earnings measured are those that appear as W2 pre-tax labor income, but we will miss contracting work and other income. For each individual we sample, we extract their age and compute their birth year from the individual characteristics file (U.S. Census Bureau (2016b)). For more details of the sample, see Abowd et al. (2009).

[1] Specifically, we sample individuals, regardless of how many years they are observed. Our sample includes data from California, Colorado, Hawaii, Idaho, Illinois, Indiana, Kansas, Maine, Maryland, Missouri, Montana, Nevada, North Dakota, Tennessee, Texas, Virginia, and Washington.

From this sample, we use the job-history file (U.S. Census Bureau (2016a)) which lists earnings at each job spell during a worker's lifetime. For each period in which the worker has earnings, we select a "dominant" employer. This is constructed in the same way as Hyatt et al. (2014). That is to say, we add earnings for the current and previous period from every employer from whom the worker has positive earnings. The dominant job is the employer with the highest two-period earnings. A job-to-job transition is a change in dominant job from one period to the next when the new dominant job had positive earnings in the same period as the old dominant job. This means that we will count job-to-job transitions in the uncommon event that a worker is continually employed at two employers but earnings fluctuations change which is the dominant job. The quarterly frequency also poses some challenges for measuring job-to-job transitions. Time aggregation implies that there may be transitions through non-employment that last up to 11 weeks. Because we require overlapping quarters of earnings, we also will miss some true job-to-job transitions that occur across a weekend that forms the seam between two quarters.

For firm-level data, we use the establishment-level quarterly-workforce indicators (U.S. Census Bureau (2016c)).[2] This is again collected at the state-level, so we use employment-weights to aggregate establishments within the state to the SEIN level. This means that multi-state firms will be counted as separate firms in each state, but transitions across establishment within firm are not counted. For each firm, we compute hires as employer-level "accessions," which includes all new employees regardless of how long the match lasts and does not exclude recalls. We trim firms that start up or shut down in that quarter and also those with more than 200% turnover.[3]

We restrict our sample in two principal ways. On the worker side, we restrict to workers who are between 25-65 years old when their job begins. Especially by cutting out the very young, we reduce labor turn-over and the prevalence of very short job-spells. On the firm side, we remove transitions to or from firms that have less than 10 employees. This again eliminates some very large volatility in terms of the measured percentage change in the firm's labor-force. The result is about 2.7 million job-to-job transitions with earnings change measured at the quarterly frequency and 1.1 million job-to-job transitions with earnings measured at the annual frequency. The reason for

[2]The firm-level data uses their entire workforce, not our 5% sample.

[3]This is possible though uncommon, given the timing of how we define turnover.

the large decline in observations of annual earnings growth is that there are many job-transitions where the origin or destination match does not last 6 quarters, which we require to measure annual earnings. This is consistent with observations in Hyatt and Spletzer (2017) who report that about $\frac{1}{3}$ of all job matches last less than 1 quarter.

1.2 Key Variables

From this dataset, we create several variables that will show up in the rest of our analysis. Our primary variable is earnings change, $\Delta w_{i,t}$. For a worker i who switches from one employer to another in period t we skip over that period's earnings to compute the growth rate

$$\Delta w_{i,t} = \ln \sum_{j=1}^{4} w_{i,t+j} - \ln \sum_{j=1}^{4} w_{i,t-j}.$$

Where $\sum_{j=1}^{4} w_{i,t+j}$ and $\sum_{j=1}^{4} w_{i,t-j}$ are full-year earnings at the origin and destination firm. Full-year earnings requires that the employer-employee match existed for 6 consecutive quarters (From t to $t+5$, and t to $t-5$), insuring that for each of the interior 4 quarters from which we calculate earnings, the worker was employed for the entire duration of the quarter. Since we do not observe when in period t the transition happened, in period t we observe earnings from two employers but do not know how many weeks of work those earnings represent. Using full-year earnings ensures that the earnings we observe cover a full year of employment. Job transitions without earnings for a full year at each firm are dropped from our sample.[4]

On the firm side, we measure job flows in several ways. Net job flows at the firm are represented by

$$\Delta L_{j,t} = \frac{H_{j,t} - S_{j,t}}{\frac{1}{2}(L_{j,t} + L_{j,t-4})}$$

where $H_{j,t}$ are the hires and $S_{j,t}$ are the separations of firm j in period t. The denominator uses the normalization of Davis et al. (1998) to bound the value between -2 and 2 while be-

[4] Using full-year earnings restricts our analysis to a subset of job transitions where we observe sufficiently long tenure in both the origin and destination job. We relax this restriction and look at quarterly frequencies in Appendix A.

ing consistent with log change to a second-order approximation. We create three more variables, used in some specifications to study non-linear effects from net job flows: net job creation, $\Delta L^{+}_{j,t} = \max(\Delta L_{j,t}, 0)$, net job destruction $\Delta L^{-}_{j,t} = -\min(\Delta L_{j,t}, 0)$, and a zero-change indicator, $\Delta L^{0}_{j,t} = \mathbb{I}_{|\Delta L_{j,t}|<0.02}$. We characterize the origin and destination firms' growth rates one quarter away from the transition period so that the growth rate of the firm is not mechanically affected by the transition of the worker we observe. That is, for a worker who transitions in quarter t, the growth rate and excess hiring rate of the origin firm, $\Delta L_{\ell,t}$ and $F_{\ell,t}$, are calculated from $t-5$ to $t-1$, and for the destination firm, $\Delta L_{d,t}$ and $F_{d,t}$ are calculated from $t+1$ to $t+5$.

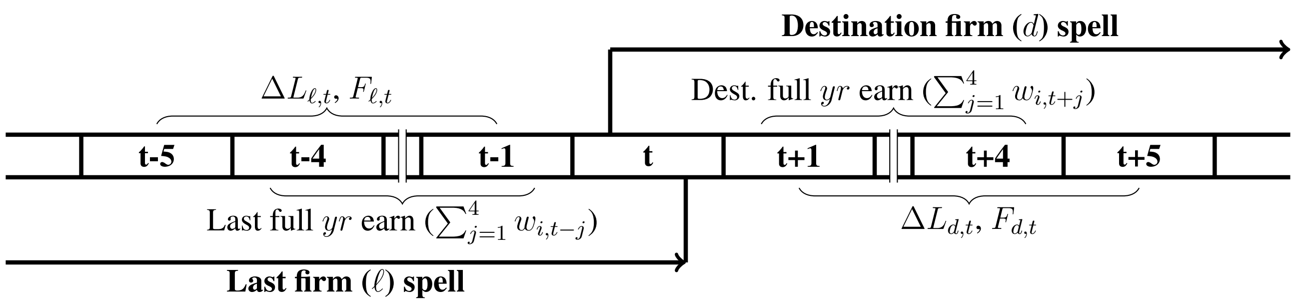

Figure 1: The timing of our employment and earnings growth measures.

To represent job "churn," the gross flows above and beyond net employment growth, we will measure excess hires. Our measure $F_{j,t}$ is the number of hires beyond that which is required to achieve the net growth at the firm. This is equivalent to the churn measure in Elsby et al. (2017) and $\frac{1}{2}$ the churn measure used in Davis and Haltiwanger (2014).

$$\begin{aligned} F_{j,t} =& H_{j,t} - \max(\Delta L_{j,t}, 0) = \min(H_{j,t}, S_{j,t}) \\ =& \frac{1}{2}\left(H_{j,t} - \max(\Delta L_{j,t}, 0) + S_{j,t} - \min(\Delta L_{j,t}, 0)\right) . \end{aligned}$$

To measure the static characteristics of workers, we use lifetime wage rank. We define $Q(\bar{w}_i|c)$, which is the quantile of total earnings across all employment and in all quarters conditional on birth cohort. [5]

[5]By conditioning on birth cohort, we control for the earnings rank of the worker relative to their position in the life-cycle earnings profile. Conditioning on year of birth also accounts for the partial measurement of workers' lifetime earnings due to the limited duration of our sample.

On the firm side, we measure firm age as the age of the firm's oldest establishment and firm size as the firm's total employment as in QWI. We measure wages of the firm as the log of the average real earnings a firm pays its employees over every quarter the firm is operational during the 15 year sample, $\tilde{w}_j$.[6] Without taking a stand on the origin of the firm effect, whether it comes from the firm itself or the workers it collects, this measure incorporates the full effect the worker may gain from the firm at which they are employed. We also control for the origin and destination firm's industry using fixed effects at the 2 digit NAICS level.

2 Unconditional Relationships

To quantify the effect of firm-level job reallocation on earnings growth, we first consider the relationship between firms' net employment growth and workers' earnings growth. Here we show that workers leaving shrinking firms, on average have less earnings growth and those going to growing firms have more earnings growth than the average job-to-job flow. Because a transition with the job flow takes a worker leaving a shrinking firm to one growing, earnings effect will be the difference between the two.

Figure 2 plots the local, unconditional correlation between earnings growth and net employment growth at the origin and destination firm for workers who make a job-to-job transition, and the relationship between net growth and earnings growth for workers who remain with their employer. We find that earnings growth in job transitions is strongly, positively correlated with the net growth of the destination firm. The origin firm's growth rate has a weaker correlation and shows lower earnings growth when workers transition away from a shrinking firm. For comparison, we also include the earnings growth of workers who stay at the same firm. This curve's level shows less earnings growth overall, but also a correlation with the firms' employment growth that is between the correlations for destination and origin. The rest of the paper will focus on earnings growth of transiters, but this stayer's slope is a useful check suggesting some passthrough in the spirit of Guiso et al. (2005). Our focus, however, is consistent with Topel and Ward (1992) and Moscarini

[6] We have also used wage rank, $Q(\tilde{w}_j)$, as the quantile of $\tilde{w}_j$. While using this regressor instead does not change the other coefficients much, its coefficient is less easy to interpret.

and Postel-Vinay (2017), who emphasize the importance of job transitions in the earnings dynamics of workers. In Appendix A, we show that these patterns hold with quarterly earnings and quarterly job flows, albeit the transiters effects are slightly steeper but the level is closer to that of the stayers, which highlights the way in which our sample of year-long tenured employees is a selected subset of all job changers.

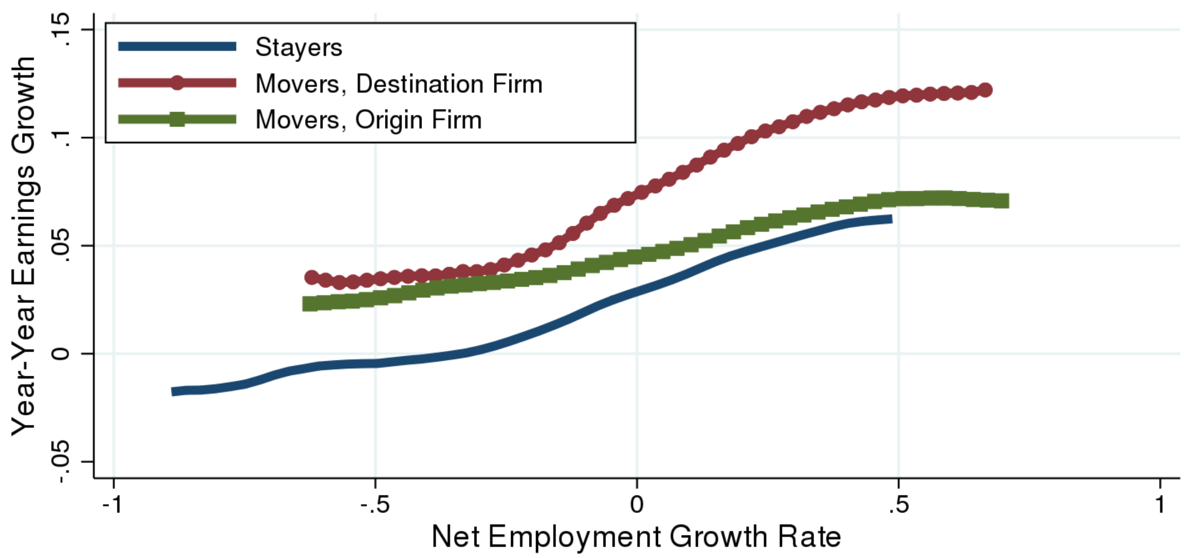

Figure 2: Nonparametric plot for earnings growth associated with net job flows at origin firm (Green), destination firm (Red) and stayers (Blue).

Not every worker going from a shrinking firm to a growing one is part of the job reallocation, because a worker going to a growing firm may actually be replacing a separation. In fact, this churn, the fraction of hires that do not contribute to net growth, is strongly correlated to net employment growth. Figure 3 shows that net employment growth and excess hires, our measure of gross "churn," are related in a V-shape. This symmetric relationship also holds at the quarterly frequency, shown in Appendix A, and indicates that as net growth increases, turnover of workers at the firm is also increasing. For growing firms, separations are also increasing and thus not every worker hired to a growing firm contributes to the firm's net job growth. Similarly, separating workers at both growing and shrinking firms are often being replaced. We consider this distinction, between worker flows that contribute to the net reallocation of jobs and the replacement hires in Section 4 when we estimate the gains associated with job reallocation.

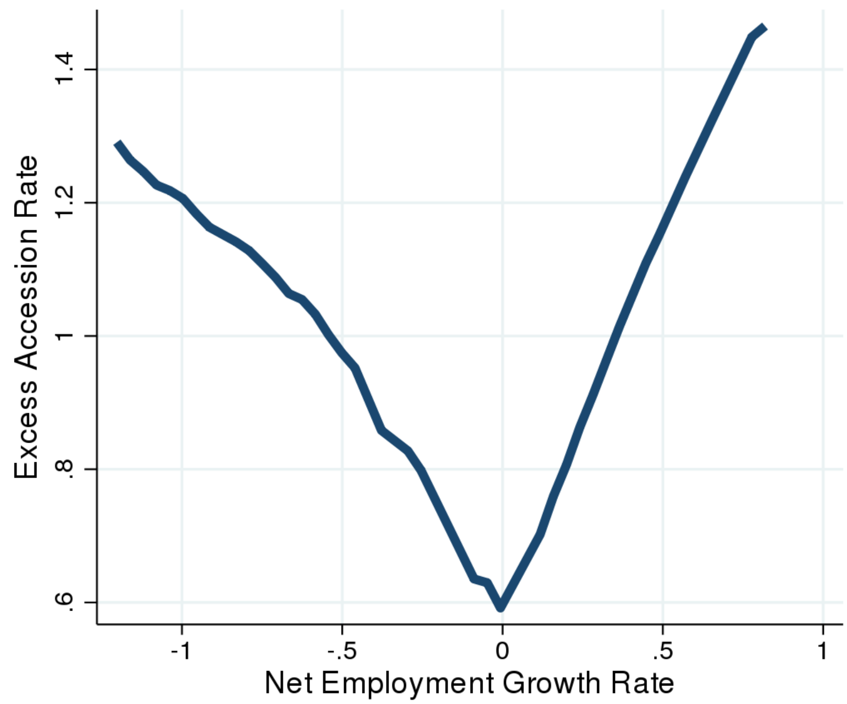

Figure 3: Nonparametric plot showing the V-shape of excess hires, churn, over net employment growth.

3 Firm-level Labor Dynamics and Individual Earnings Growth: Conditional Results

Many firm characteristics play into workers' earnings growth during transitions, and so in this section we will check that the patterns in Section 2 remain after conditioning on other, known determinants and compare their magnitudes to that of firm labor dynamics. In this section, we present our primary reduced form estimates of earnings growth with the origin and destination firms' characteristics. We estimate the following regression specification:

$$\begin{aligned}\Delta w_{i,t} =& \beta_{1,d}\Delta L_{d,t} + \beta_{1,\ell}\Delta L_{\ell,t} + \beta_{2,d}F_{d,t} + \beta_{2,\ell}F_{\ell,t} \\ &+ \beta_{3,d}\tilde{w}_d + \beta_{3,\ell}\tilde{w}_\ell + \beta_4 Q(\bar{w}_i|c) + x'_{\ell,d}\beta\end{aligned} \tag{3.1}$$

where $x'_{\ell,d}$ denotes origin and destination log employer size, employer age, a quadratic in worker age, gender, 2-digit SIC industry dummies and time effects for the quarter of the transition. The indices for the origin, ℓ, and destination, d, are functions of i, t but we suppress notation for this. Results for the regression in Equation 3.1 are presented in Table 1. The first column is the baseline specification. The second breaks net employment growth, ΔL_j into job creation and destruction, ΔL_j^+ and ΔL_j^-. Column (3) looks at the same specification but with worker-level fixed effects. Column (4) includes the accession rate, H, in lieu of the excess accession rate F. All of the coefficients except firm age at the destination firm in Column (3) are significant at the 99% level.

The first two coefficients, net employment growth at the origin and destination firm are most interesting to us. Both are consistent with the positive slopes we showed in Figure 2. While both moving from and going to a net growing firm brings a positive earnings gain, the elasticity is much larger at the destination firm. The coefficients in Column(1) mask the asymmetry between net job creation and destruction. Column (2) splits ΔL_j in two, between job creation and job destruction, $\Delta L_j^+, \Delta L_j^-$ at the origin and destination $j \in \{\ell, d\}$. The largest coefficients are on job creation at the destination, ΔL_d^+ and job destruction at the origin, ΔL_ℓ^-. workers who are hired at a growing firm have greater earnings growth than otherwise expected, with an elasticity of 10.7%. Workers who leave shrinking firms see their wages decline relative to that which is otherwise expected. The elasticity of earnings growth to job destruction at the origin is about -7.3%. Note that job creation at the origin firm has little positive effect on earnings growth.

Column (3) adds individual-level fixed effects. Meaning that we are now looking within an individual's labor market history comparing earnings growth when the same individual transits to e.g. a growing firm or a shrinking firm. While differencing earnings removed individual effects from earnings levels this removes individual trends. Further, this is one way of addressing selection on unobservable characteristics, i.e. some workers may be more likely to switch to growing firms and more likely to have earnings growth in the process. Column (3) shows our results are not driven by this potential composition effect.

Column (4) is particularly important because it clarifies why we focus on net employment growth rather than simply hires. Comparing coefficients, which are comparable elasticities, there is an empirical distinction: employment growth itself is more important than hires or the turnover coefficients F. Essentially the question here is whether what matters for earnings growth is hiring or net growth/creation? The coefficient on the accession rate is positive but much smaller than the net flow rate. To paraphrase much prior work that distinguishes gross hires from neg growth (e.g. Elsby et al. (2017), Column (4) for helps differentiate whether earnings growth primarily reflects recruitment itself, which would mean the accession rate is quantitatively largest, or if the earnings growth premium is driven by firms trying to increase their labor input, implying employment growth itself is largest. The result motivates our exercise below, which estimates the probability a hire is part of the net employment growth and not part of the excess turnover.

Finally, Table 1 also allows us to quantitatively the relationship of earnings growth and firm dynamics to other firm static characteristics known to be important such as size, age, and average wage. The wage level of the firm, $\tilde{w}_x$ is large, about twice the size of employment growth, ΔL_d and four times the size of ΔL_ℓ. But, the magnitudes of all the coefficients are very small. We see earnings growth coming from workers climbing a job-ladder, and while many of the other characteristics, size and F_x, for instance, they are mostly subsumed by the effect of going from lower- to higher-wage firms. ΔL_x, however, still has a large effect on earnings growth, even after conditioning on the workers' movement in the average-wage ladder.

To compare these implied effects in dollar values, we use the estimates from Table 1 and the transition patterns from the data to estimate the 90th percentile most earnings growth associated with each characteristic and the 10th percentile most earnings loss—that is, the dispersion in earnings growth across job-to-job transitions that can be ascribed to each firm characteristic. The difference in firms' average wages implies a 90-10 differential of $\$7,578$, the difference between firm sizes explains $\$2,250$ and $\Delta L_\ell, \Delta L_d$ implies a 90-10 differential of $\$1,624$. This is to say, whether a worker moves from a low- to a high-wage firm is the largest predictor of the dispersion in earnings growth during a job-to-job transition. But moving up the firm size distribution and moving towards growing firms have effects of approximately the same scale, 20-30% of the average wage effect.

4 Quantifying the Contribution of Net Job Flows

In this section, we will present our estimates to answer our initial question: How does moving with the job flow contribute to earnings growth among job-to-job transitions? To revisit the logic of our answer, Section 2 presented the first part: going to a growing firm implies more-than-average earnings growth, leaving a shrinking implies less-than-average and thus going with the job flow subtracts the former from the latter. However, Figure 3 showed that growing and shrinking firms have considerable turnover, so a hire at a growing firm is actually likely to be a replacement hire, rather than net growth. Table 1 showed this distinction matters because net growth measures had significantly larger earnings growth elasticities than excess hires or total hires.

We cannot directly observe whether a particular hire at a growing firm replaced a separation or is part of net employment growth, and similarly, we cannot tell if a separator from a shrinking firm was replaced. However, we can estimate the probability that a worker was part of the job flow across firms based on the origin and destination firms' net and gross flows.

To understand how we calculate this probability, consider a worker who moves from a stagnant origin firm, $\Delta L_\ell = 0$, to a firm with a net increase of one worker and a total hiring of 2 workers meaning there was 1 separation. This worker had a 50% chance of being a net-growth hire and a 50% chance of being a replacement. We can treat the origin analogously, as a firm separating 2 but also hiring 1, shrunk by 1 and each separation had a 50% chance of being part of the net job flow. We then calculate the probability of being part of the job flow, $\mathrm{Pr}[JF]$, contributing to the net reallocation of workers between firms, as follows:

$$\mathrm{Pr}[JF] = n_d n_\ell + n_d \frac{F_\ell}{S_\ell} + n_\ell \frac{F_d}{H_d} \tag{4.1}$$

where $n_d = \frac{\Delta L_d^+}{H_d}$ and $n_\ell = \frac{\Delta L_\ell^-}{S_\ell}$, using H_d, S_ℓ to be the rates of hires and separations. These are the probabilities that a hire or separation is part of the net flow from a shrinking or towards a growing firm. The other terms, $\frac{F_\ell}{S_\ell}$ and $\frac{F_d}{H_d}$ represent the complement, that a hire at a growing firm was a replacement. To put this whole probability into words, $n_d n_\ell$ is the chance the transition was both a net growth at the destination and net shrink at the origin, while $n_d \frac{F_\ell}{S_\ell}$ and $n_\ell \frac{F_d}{H_d}$ are the chance of a net growth at the destination and replaced loss at the origin or net loss at origin and replacement hire at the destination. The first and second terms are the probability of being part of job reallocation towards a growing firm and the first and third terms are the probability of job reallocation away from a shrinking firm. This is an inclusive notion of being part of net job reallocation because a worker can contribute by moving into a faster growing firm or by leaving a shrinking firm.

To quantify the earnings growth associated with $\Pr[JF]$, we run our baseline regression specification, but replace other firm dynamics terms, ΔL_x, F_x, H_x, with $\Pr[JF]$. These results are in Column (1) of Table 2. We then break apart the effect of going towards a growing firm and leaving a shrinking firm, $n_d n_\ell + n_d \frac{F_\ell}{S_\ell}$ or $n_d n_\ell + n_\ell \frac{F_d}{H_d}$, in Column (2). These confirm the logic from earlier, workers hired for firm growth experience above-average earnings growth, and those leaving as part of firm shrinking experience below-average earnings growth. The former is stronger than the latter, so their net effect is positive. use several measures which indicate that a job transition is part of the net reallocation of workers, moving with the job flow. Columns (3) and (4) use coarser measures as a baseline check. In Column (3), we replace $\Pr[JF]$ with an indicator $\mathbb{I}_{\Delta L_d - \Delta L_\ell > 4\%}$, another possible notion of job flows but which does not control for the rate of excess hires.[7]

From these estimates, we can conclude that workers moving with the job flow reap a considerable earnings benefit. The coefficient on $\Pr[JF]$ in Column (1) suggests that a transition with a 100% chance of being with the job flow gets 1.7 percentage points more earnings growth than a job transition that is certainly part of the worker churn. This is 23% of the earnings increase of an average transition, 7.3%. Again, the positive effect from transiting with the job flow comes from the effect of growth at the destination firm. Column (2) suggests that a worker who contributes to employment growth at the destination has 9.1 percentage points more earnings growth, more than

[7] An alternative construction, $\mathbb{I}_{\Delta L_d > 2\% \cap \Delta L_\ell < -2\%}$ is quantitatively quite similar.

double the earnings growth of an average transition.

Of course, the missing piece is how frequently workers move with the job flow. About 46% of hires are to firms growing more than 2% in a year while 34% of hires go to firms *shrinking* more than 2%. Separations come from shrinking firms 37% of the time, and from growing firms 43% of the time. Of course, Figure 3 suggests that many of these hires do not contribute to net employment growth. We find 20.7% of hires contribute to net job flows, the sample average of our $\Pr[JF]$ measure. This implies that the average transition gets $\$183$ from the premium associated with moving with a job flow—the 0.017 coefficient times 0.207 and converted to dollars. This effect is composed of a $\$516$ premium from workers going to growing firms, but a $\$243$ loss from leaving a shrinking firm.

5 Discussion/Conclusion

In this paper, we show that firms' employment dynamics are related to the earnings growth that workers receive when they switch jobs. Workers' earnings growth is particularly responsive to employment growth at the destination firm and employment declines at the origin firm. Furthermore, net employment growth is considerably more important than gross hires. Combining these finds, we estimate how moving with the cross-firm job flow affects earnings growth for job-to-job transitions.

We see these findings as informative for the relationship between firm dynamics and wages empirically, but also as testable implications for a range of theoretical models of the labor market. For theories of firm-risk sharing, job transitions are a quantitatively important facet of the effects of firm-level shocks on workers. While workers who stay with an employer see their earnings growth change with their employers' dynamics, much of the effects from these dynamics are on the earnings growth of job transitioners who are leaving or coming into the firm.

Search models of the labor market where firms employ multiple workers are a largely recent development. However, these models provide an ideal framework for understanding the relationship between wage growth and firm growth. Our empirical results are ideal targets for calibrating such models to produce counterfactual exercises and highlight the economic mechanisms through which this relationship operates.

Our findings linking and distinguishing gross and net flows emphasize the importance of this work, on firms in worker job search matching. Our facts, that turnover and employment growth are correlated at firms, with a V-shape, yet have dissimilar elasticities for earnings growth, give insight into how firms use the promise of earnings to grow, rather than just to hire. They also give an important dimension by which job transitions can be distinguished. These differences between net growth and turnover in our earnings growth regressions also necessitates our probabilistic approach to isolate the effect of reallocation.

Labor market search models are clear about the importance of the origin firm's role for understanding the incentives of a worker to set their reservation wage or search for a particular increase in their present value via on-the-job-search. However, the theory is less clear about what determines the magnitude of earnings growth and whether this should depend on the origin or destination firm. In Delacroix and Shi (2006), workers search for incrementally lower gains in wages as they move up the job ladder and face lower probabilities of job-finding. This is consistent with our finding that earnings growth is decreasing in the average wage of the origin firm. However, we also find that earnings growth increases with the wage level of the destination firm and Delacroix and Shi (2006) lacks a concept of firm size. In terms of earnings growth and net employment growth, Coles and Mortensen (2016) provides a possible framework for thinking about firm growth and earnings growth via on-the-job search. [8]

This empirical feature, that the hiring firm matters, and that being hired at a growing firm has particular importance for earnings growth (i.e. accounts for nearly 10% of earnings growth) has particularly interesting implications when considering fluctuations in firm dynamics. While the analysis was done without time interactions, it suggests that when there are fewer fast growing firms, whether because of a cyclical downturn or the long-trend decline in entry, this may have ancillary effects on earnings growth of job-changers and thus earnings growth overall.

[8] Information may also play a role given the importance of the destination employer. For instance, if firms cannot credibly report their productivity growth (both growing and shrinking firms look the same to searchers), then wages may be set independently of the firm's growth rate and it may appear as a positive earnings shock which accrues to otherwise identical workers after matches are made.

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Table 1: Baseline Estimates of Earnings Growth across Job-to-Job Transitions.

	(1)	(2)	(3)	(4)
ΔL_ℓ	0.0280		0.0437	0.0321
	(0.00144)		(0.00290)	(0.00151)
ΔL_d	0.0717		0.0769	0.0651
	(0.0033)		(0.0023)	(0.00145)
ΔL_ℓ^+		-0.00961		
		(0.00221)		
ΔL_d^+		0.107		
		(0.00236)		
ΔL_ℓ^-		-0.0731		
		(0.00246)		
ΔL_d^-		-0.0412		
		(0.00214)		
H_ℓ				-0.00659
				(0.00063)
H_d				0.0141
				(0.00071)
F_ℓ	-0.00386	-0.00210	-0.00214	
	(0.00066)	(0.00066)	(0.00179)	
F_d	0.0117	0.00947	0.0181	
	(0.00075)	(0.00076)	(0.0021)	
$ln(Emp_\ell)$	-0.00429	-0.00458	-0.00404	-0.00429
	(0.00015)	(0.00015)	(0.00036)	(0.00015)
$ln(Emp_d)$	0.00158	0.00186	0.00175	0.00162
	(0.00014)	(0.00015)	(0.00036)	(0.00014)
$Firm\ Age_\ell$	0.00047	0.00041	0.00045	0.00045
	(0.00004)	(0.00004)	(0.00009)	(0.00004)
$Firm\ Age_d$	-0.00021	-0.00016	-0.00008	-0.00018
	(0.00004)	(0.00004)	(0.00008)	(0.00004)
$\tilde{w}_\ell$	-0.143	-0.143	-0.120	-0.145
	(0.00083)	(0.00083)	(0.00216)	(0.00082)
$\tilde{w}_d$	0.157	0.156	0.130	0.158
	(0.00083)	(0.00083)	(0.00229)	(0.00083)
Constant	0.192	0.195	0.615	0.193
	(0.00986)	(0.00985)	(0.0516)	(0.00982)
N	1054000	1054000	476000	1054000
R^2	0.105	0.105	0.0789	0.105

Includes controls for worker age and its quadratic, age gender, lifetime earnings rank conditional on birth cohort, industry and time effects. Standard errors in parentheses. All columns present annual earnings growth. Column 3 contains the individual fixed effects estimation. Column 4 includes the accession rate, H, in lieu of the excess accession rate F.

Table 2: Estimates of Earnings Growth due to Job Flows in Job-to-Job Transitions.

	(1)	(2)	(3)
$\Pr[JF]$	0.017		
	(0.00148)		
$n_d n_\ell + n_d \frac{F_\ell}{S_\ell}$		0.0906	
		(0.00195)	
$n_d n_\ell + n_\ell \frac{F_d}{H_d}$		-0.0418	
		(0.00171)	
$\mathbb{I}_{\Delta L_d - \Delta L_\ell > 4\%}$			0.0123
			(0.0006)
$Q(\bar{w}_i \| c)$	0.0166	0.0157	0.0166
	(0.00183)	(0.00183)	(0.00183)
$ln(Emp_\ell)$	-0.00436	-0.00447	-0.00436
	(0.00015)	(0.00015)	(0.00015)
$ln(Emp_d)$	0.00148	0.00207	0.00151
	(0.00036)	(0.00014)	(0.00014)
$Firm\ Age_\ell$	-0.011	-0.0109	-0.011
	(0.0002)	(0.0002)	(0.0002)
$Firm\ Age_d$	9E-05	9E-05	9E-05
	(0.000002)	(0.000002)	(0.000002)
$\tilde{w}_\ell$	-0.142	-0.139	-0.141
	(0.00073)	(0.00073)	(0.00072)
$\tilde{w}_d$	0.15	0.146	0.151
	(0.00074)	(0.00074)	(0.00074)
Constant	0.248	0.255	0.233
	(0.00882)	(0.00881	(0.00877)
N	1054000	1054000	1054000
R^2	0.102	0.104	0.102

Includes controls for worker age and its quadratic, age gender, lifetime earnings percentile conditional on birth cohort, industry and time effects. Standard errors in parentheses. All columns present annual earnings growth.

Table 3: Earnings Growth from Reallocation vs. Average Job Transition.

	β	Flow Probability	Job Flow Effect	$\mathbb{E}(\Delta w)$
$\Pr[JF]$	0.017	0.207	\$ 183.6	\$ 2921
$n_d n_\ell + n_d \frac{F_\ell}{S_\ell}$	0.091	0.111	\$ 515.8	\$ 2921
$n_d n_\ell + n_\ell \frac{F_d}{H_d}$	-0.042	0.109	\$ -242.9	\$ 2921

Column (1) shows the estimated average effect of job flows, Column (2) the sample average of $\Pr[JF]$, Column (3) multiplies (1) and (2) and converts that into dollar terms and column (4) lists the average earnings growth in our sample. The second and third rows break JF into flows at the destination and origin.

Online Appendix

A Additional Figures

A.1 Quarterly frequency

In this section we present quarterly frequency estimates for Figures 2 and 3. Earnings growth is slightly lower among the job movers, which reflects both selection and job tenure effects. In the quarterly sample we have many short-tenured workers who do not last a full year at their two matches and these workers tend to gain less in job-to-job transitions.

In Figure 5, we see the same V-shaped relationship between excess hires and the net employment growth as in Figure 3. Note that at the quarterly frequency, the excess accession rate, the churn rate, is much lower than at the annual frequency. This is almost by construction because the total number of hires accumulates over a longer period.

A.2 Job Flow Probability

In Table 4 we show the probabilities that a given job transition in our sample was a move from a growing, shrinking, or stagnant ($< 4\%$ absolute change in employment) firm to a growing, shrinking or stagnant destination firm. The sum of all cells is 1. While the plurality of transitions are to growing destination firms, there are substantial numbers of job transitions to shrinking and stagnant firms. To identify what worker flows are associated with a job flow, we not only have to consider the movement of workers into growing firms and away from shrinking ones, but also the gross flows and churn of these associated firms.

A.3 Distribution of Earnings Gains

To ilustrate the importance of some firm-level variables in terms of their implied effect on earnings growth, we look at the earnings differentials implied by their regression coefficients at different

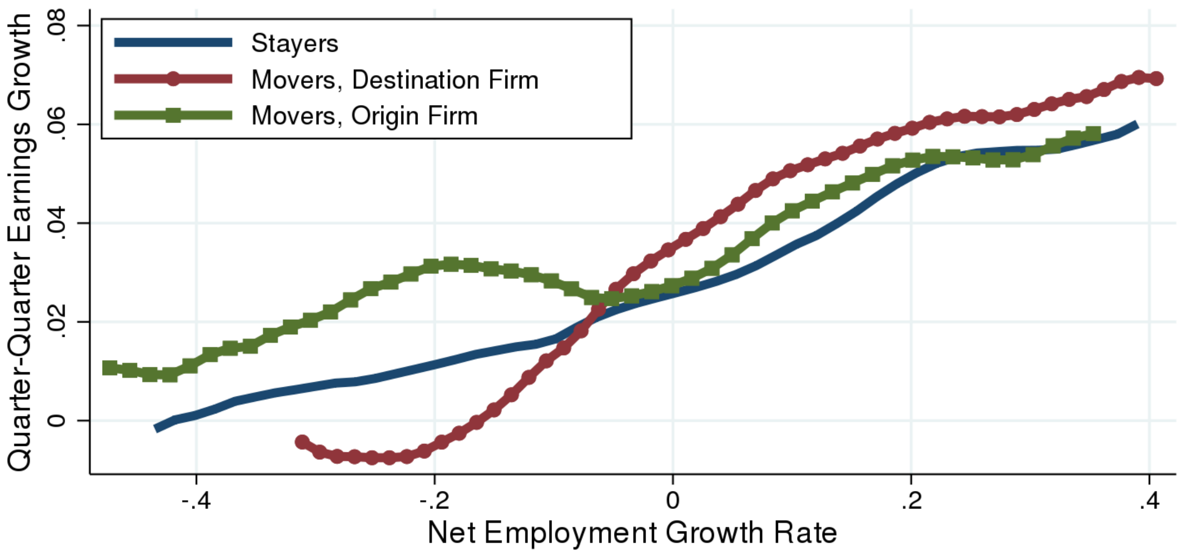

Figure 4: Quarterly frequency, nonparametric plot for earnings growth associated with net job flows at origin firm (Green), destination firm (Red), and stayers (Blue).

Table 4: Probability matrix of Job Transition Types

	Destination		
Origin	Grow	NoChng	Shrink
Growth	0.22	0.08	0.13
NoChng	0.08	0.05	0.06
Shrink	0.16	0.07	0.15

percentiles. We take the dollar amount implied by the coefficient on the variable in our baseline regression and multiply it by the difference between the following percentiles of that variable: 75-25, 90-50, 50-10, and 90-10. The difference in earnings growth across the distribution of these firm characteristics is large.

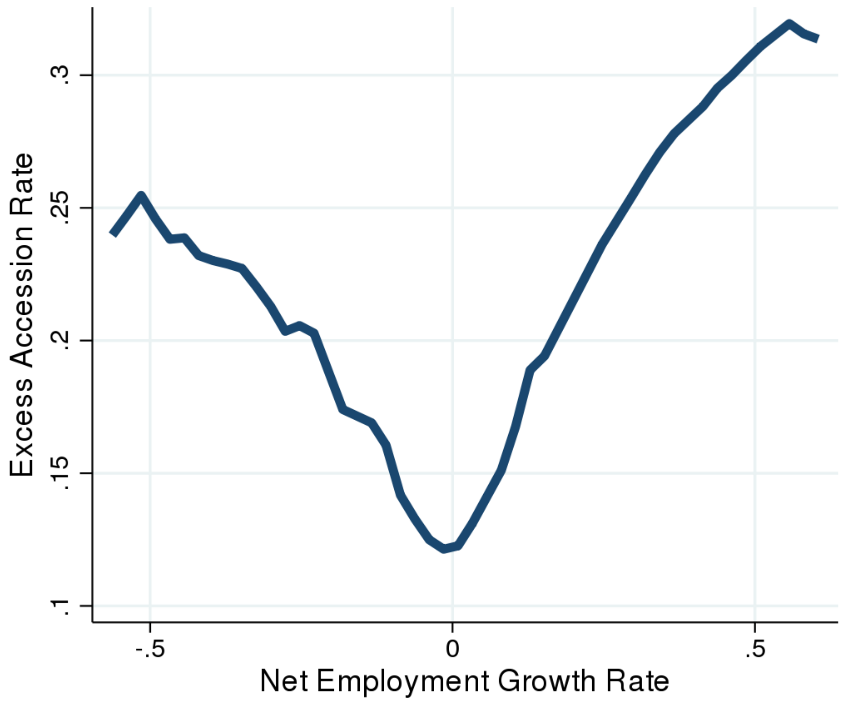

Figure 5: Quarterly frequency, nonparametric plot showing the V-shape of excess hires, churn, over net employment growth.

Table 5: Distribution of Implied Earnings Gains

Variable	p75-p25	p90-p50	p50-p10	p90-p10
ΔL	453.9	853.3	770.7	1624
ΔL_d	398.8	1022.4	332.7	1355.1
ΔL_ℓ	192.7	50.2	564.5	614.7
$\tilde{w}$	3442	5589	1989.4	7578.4
$ln(Emp)$	1037.8	590	1661.6	2251.6